

The Effects of Tension and Contractility on Cell Packing Geometry

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Abstract

The packing geometry of confluent epithelial tissue affects many developmental processes. A two-dimensional apical vertex model is useful for examining the behaviour of such tissues. With this form of modelling, we gain insight into phenomena such as Lewis's law and tissue stability. In this paper we expand on the results from Farhadifar et al. (2007) [1]. We also apply their analysis to a variation of the Farhadifar vertex model, and expose trends in Lewis's law as the parameters in each model vary. We end with a discussion on potential improvements to our method and avenues for future research.

1 Introduction

1.1 Motivation

The packing geometry of cells is vital to many developmental processes in living organisms. It is especially important during early stages of growth such as the embryonic stage in mammals. Packing geometry affects the structure and positioning of the first cells in an organism. This can determine organ placement and other vital characteristics during the early stages of growth [2]. A key geometric phenomenon of cells is *cell sorting*. Cells have been observed sorting themselves into groups by varying the adhesion of individual cells to each other. The concept of adhesion is further discussed in Section 1.2. Cell sorting is also strongly related to the growth of cancerous cells. This makes it a key area of interest for various cancer research organisations such as Cancer Research UK [3].

Many key processes also result from forces in just the *epithelial*, or surface layer of the tissue. The exposed face of this layer is known as the *apical* layer [3]. The epithelial layer is also *confluent*, meaning there is no space between cells. The formation of hollow cylindrical tubes in an organism can be attributed to the geometry of an epithelial layer. An example of this tubing is the neural tube found in vertebrates [4]. Another widely studied epithelial phenomenon occurs in the *drosophila* or fruit fly wing. The epithelial tissue in the wing of a fruit fly is typically disordered and contains cells of various shapes and sizes. However, prior to hair growth, the tissue forms a regular hexagonal lattice [1]. Epithelial tissue modelling is useful as modelling the apical layer as a two-dimensional lattice is much simpler than a three-dimensional system. One advantage to examining epithelial tissue is its simplicity relative to three-dimensional clusters of cells. Despite its simplicity, it still provides reliable insight into cell dynamics [3].

1.2 The Biology of Cell Packing

Cell tissue was previously believed to behave as a foam or froth. In such a model, the shape and size of each cell is determined by surface tension and internal pressures [2]. Like a bubble in a foam, or a drop of oil in water, the surface area of a cell is minimised. Each cell reaches an equilibrium between its internal pressure and this surface tension; this determines the packing geometry. This is a good approximation for cell clusters. In fact, cell sorting can be explained with just this simple model [2]. However, we now know that there are various *active* cell components that also contribute to this equilibrium state. These components are collectively referred to as the *actomyosin* complex [5]. This complex contains actin filaments and myosin motors which induce contractile forces on the cell membrane. Furthermore, we must also account for cell adhesion which is the force between cells which allow them to attach to one another. This process is mediated by the *cadherin* molecule. Thus, a versatile mathematical model which can describe these properties is required.

1.3 Vertex Models

A common method for modelling confluent epithelial tissue is a *vertex model*. In a vertex model, each cell is represented by a polygon in two-dimensional space. We refer to the sides of each polygon as edges and the corners of the polygons as vertices. An epithelial layer is a collection of vertices connected by edges to form a continuous sheet of neighbouring cells. In reality, epithelial sheets are three-dimensional. Therefore, we interpret these two-dimensional vertex models as just the apical layer of the sheet [4]. It is known that the actomyosin machinery in a cell lies just beneath the apical layer [1]. This justifies the usage of our two-dimensional approximation.

We perform simulations on these sheets by modelling the forces exerted on the cell by formulating models of forces on each vertex in the sheet. To define these forces, we first define a potential energy function $E(\mathbf{R}_i)$ where $\mathbf{R}_i$ is the position of vertex i . It is common to index vertices with i or j and cells with α . We then refer to edges by the two vertices they join, such as $\langle i, j \rangle$.

The force on a given vertex is the negative gradient of the potential energy, that is,

$$\mathbf{F}_i = -\nabla E(\mathbf{R}_i). \quad (1)$$

We equate these forces with some friction force $\eta \mathbf{v}_i$, where $\mathbf{v}_i$ is the velocity of vertex i [6]. Little is known about the frictional forces inside a tissue, so we apply this simple model for a drag force which is proportional to the velocity of the vertices [3]. This yields the first order differential equation

$$\eta \frac{d\mathbf{x}_i}{dt} = \mathbf{F}_i. \quad (2)$$

Upon solving equation (2), we obtain the position of the vertices as the sheet relaxes from an unstable arrangement to a stable equilibrium state. Over time, the system will approach a stable, attracting fixed point. Here, the net force on each vertex is zero and the system will remain in a constant state unless disturbed or altered by external factors. This formulation for vertex modelling can be applied to foams and cells. The differences between modelling these systems come from the specific energy functions we define later.

1.4 Lewis's Law

Lewis's law was first discovered in 1928 by Frederic T. Lewis. It states that in an epithelial tissue, the mean area of the set of all n sided cells is linear with respect to n [7]. Lewis discovered this law after observing cucumber cells under a microscope. In more recent years, the results have been replicated in computer simulations of cell systems.

A notable result is that of Farhadifar et al. in 2007 [1]. They used a specific vertex model, now known as the Farhadifar model. This model introduced a perimeter elasticity term to other commonly used vertex models which is further discussed in Section 2.1. With this change, the simulated sheets conformed to Lewis's law, which is vital for any model attempting to match real world observations. Furthermore, they showed that the gradient of this linear relationship is dependent on the nondimensionalised parameters of the model.

1.5 Outline

We will first define the vertex models analysed in this report in Section 2. We will then recreate the results from Farhadifar et al. (2007) [1] in Section 3.1. Here, we will also quantitatively describe the effect of the model parameters on the gradient of Lewis's law. We will perform the same analysis on an adjusted model using a line elasticity model in Section 3.2. This is a common variation of the Farhadifar model, notably used in Noll, Nicholas, et al. (2017) [8]. Finally, we will end with a discussion of the results and future potential research in Section 4.

2 Models and Method

2.1 The Farhadifar Model

The first specific cell-vertex model used in this report is that of Farhadifar et al. [1]. The energy function consists of a quadratic area term, a linear line tension term and a quadratic perimeter term. The energy, E , as a function of the position of a vertex i denoted by $\mathbf{R}_i$ is therefore

$$E(\mathbf{R}_i) = \sum_{\alpha} \frac{K}{2} (A_{\alpha} - A_0)^2 + \sum_j \Lambda \ell_{i,j} + \sum_{\alpha} \frac{\Gamma}{2} P_{\alpha}^2. \quad (3)$$

In this equation, K , Λ and Γ are scalar parameters which determine how much each component of the energy function affects the total energy. These are called the area elasticity, line tension, and contractility, respectively. The preferred area is represented by A_0 . Cells with an area A_0 experience no forces from the area elasticity. Furthermore, if we ignore the line tension and perimeter contractility forces, cells with an area greater than A_0 contract. Conversely, cells with an area less than A_0 expand. We denote the area and perimeter of a cell α with A_{α} and P_{α} respectively. Considering only the line tension term, if the line tension Λ is positive, the edges contract. A negative line tension causes the edges to expand. The contractility Γ is non-negative, and so the perimeter contractility term always induces a cell contraction.

The area elasticity term represents the forces from the internal pressure of the cell. The perimeter elasticity term represents the actomyosin network exerting active contractions on the cell boundary. The actomyosin complex is only capable of producing contracting forces. For this reason, we require that the contractility Γ is non-negative. Conversely, the line tension represents a combination of the surface tension, actomyosin proteins, and cell adhesion. The surface tension and actomyosin network produce contraction forces on the edges. Adhesion produces expansion forces. This is why we allow the line tension Λ to be positive or negative; the sign depends on the relative sizes of these three factors.

Note that the actomyosin network contributes to two terms in the energy function. As we will see in Section 2.3, there is no agreed upon method for modelling the forces from the actomyosin system. These forces can be approximated by a combination of linear and quadratic terms in the energy function [1]. Yet still, there are various ways of doing this.

We can use the chain rule to compute the gradient of equation (3) and find the forces on each vertex using the Farhadifar model. We obtain

$$\nabla E(\mathbf{R}_i) = \sum_{\alpha} K(A_{\alpha} - A_0)(\nabla A_{\alpha}) + \sum_j \Lambda(\nabla \ell_{i,j}) + \sum_{\alpha} \Gamma P_{\alpha}(\nabla P_{\alpha}).$$

The detailed derivation of the force directions can be found in Appendix A. We find the total force on each vertex to be

$$\mathbf{F}_i = \sum_{\alpha} \frac{K}{2} b(A_{\alpha} - A_0) \hat{\mathbf{n}}_b + \sum_j \Lambda \hat{\ell}_{i,j} + \sum_{\alpha} \Gamma P_{\alpha}(\hat{\ell}_{i,1} + \hat{\ell}_{i,2}). \quad (4)$$

The directions of the unit vectors in Equation (4) are shown in Figure 1.

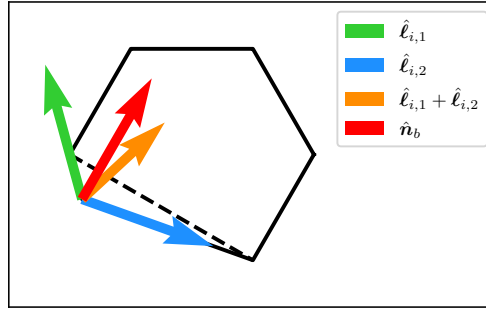


Figure 1: The direction of each of the force components in equation (4) for a single cell.

2.2 Nondimensionalisation

A key result from Farhadifar et al. (2007) is that the gradient of the linear Lewis's law relationship depends directly on the parameters of the model. They express this notion in terms of two nondimensionalised parameters, $\bar{\Lambda}$ and $\bar{\Gamma}$. As we will see, the characteristics of the cell sheet that we are interested in are only dependent on two independent combinations of the four dimensional parameters. We see that nondimensionalising the model is a useful step because it reduces the number of parameters we need to analyse. Consequently, it reduces the number of simulations which need to be computed. This is true in general when nondimensionalising a system.

To obtain the nondimensionalised form of equation (3) we first define a characteristic length $\ell_0 = \sqrt{A_0}$. Upon replacing spatial quantities with their size relative to ℓ_0 , we find that equation (3) becomes

$$E(\mathbf{R}_i) = \sum_{\alpha} \frac{K}{2} (\bar{A}_{\alpha} \ell_0^2 - \ell_0^2)^2 + \sum_j \Lambda \bar{\ell}_{i,j} \ell_0 + \sum_{\alpha} \frac{\Gamma}{2} (\bar{P}_{\alpha} \ell_0)^2.$$

Note that the variables with a bar above them represent the dimensionless quantities. Nondimensionalising the energy, we then obtain

$$\bar{E}(\mathbf{R}_i) = \sum_{\alpha} \frac{K \ell_0^4}{2 E_0} (\bar{A}_{\alpha} - 1)^2 + \sum_j \frac{\Lambda \ell_0}{E_0} \bar{\ell}_{i,j} + \sum_{\alpha} \frac{\Gamma \ell_0^2}{2 E_0} \bar{P}_{\alpha}^2,$$

where E_0 is a characteristic energy quantity that is yet to be defined. Notice that $E_0 = K \ell_0^4$ is a convenient choice as the area elasticity term no longer depends on any parameters. Physically, $K \ell_0^4$ represents twice the

energy required to fully contract and collapse a cell of area A_0 . Now, rewriting this with $A_0 = \ell_0^2$ gives the fully nondimensionalised form of the energy equation. That is,

$$\bar{E}(\mathbf{R}_i) = \sum_{\alpha} \frac{1}{2} (\bar{A}_{\alpha} - 1)^2 + \sum_j \bar{\Lambda} \bar{\ell}_{i,j} + \sum_{\alpha} \frac{\bar{\Gamma}}{2} \bar{P}_{\alpha}^2, \quad (5)$$

where

$$\bar{\Lambda} = \frac{\Lambda}{K A_0^{3/2}} \quad \text{and} \quad \bar{\Gamma} = \frac{\Gamma}{K A_0}. \quad (6)$$

These definitions are the same as those used in Farhadifar et al. (2007) [1]. The nondimensionalised forces can be computed with these definitions also.

It is important to note that only certain pairs of parameters $\bar{\Lambda}$ and $\bar{\Gamma}$ produce stable tissues. If the surface tension and contractility of the cells are too large, the area elasticity term will not be large enough to prevent the sheet from collapsing. In their 2007 paper, Farhadifar et al. produced a phase transition diagram illustrating this fact [1]. This is shown in Figure 2.

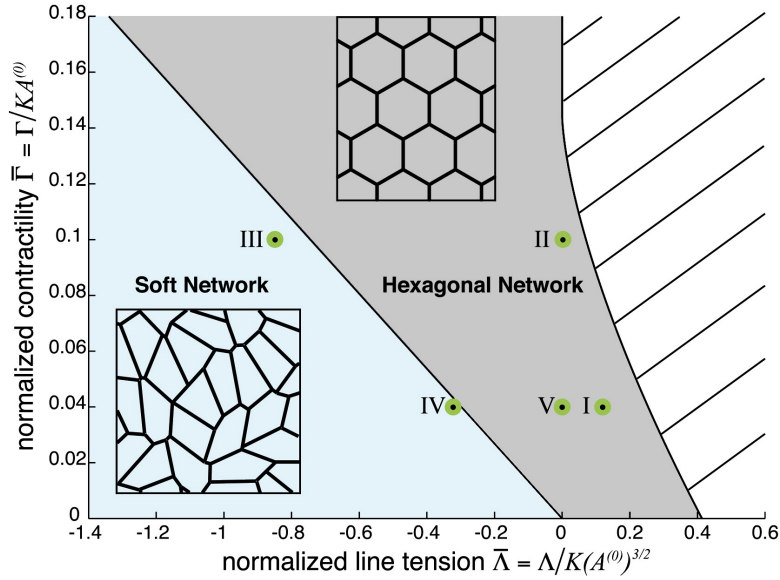


Figure 2: The phase plane for the nondimensionalised Farhadifar Model as published by Farhadifar et al. [1]. The blue region represents a fluid-like state in which the area elasticity term dominates, and the cell shape is less relevant. The grey region represents a state in which a hexagonal arrangement is the global energy minimum. This forms a more rigid structure. The invalid region on the right represents a state where the contractile forces are too great, and the sheet collapses on itself [1].

2.3 A Line Elasticity Model

We now investigate a variation of the Farhadifar model in which the perimeter contractility term is replaced with a line elasticity term. In doing this, we change our model for the active contractions in the actomyosin network. This term in the energy function is given by

$$E_{\Omega}(\mathbf{R}_i) = \sum_j \frac{\Omega}{2} \ell_{i,j}^2 \quad (7)$$

where Ω is a line elasticity coefficient. The subtle difference here is that the contractile force along an edge depends only on the length of that edge, not the perimeter of the cell. This model has been previously investigated by Noll, Nicholas, et al. (2017) [8].

Nondimensionalising this term is similar to nondimensionalising the perimeter elasticity term. The nondimensionalised energy function for the new model is therefore

$$\bar{E}(\mathbf{R}_i) = \sum_{\alpha} \frac{1}{2} (\bar{A}_{\alpha} - 1)^2 + \sum_j \bar{\Lambda} \bar{\ell}_{i,j} + \sum_j \frac{\bar{\Omega}}{2} \bar{\ell}_{i,j}^2. \quad (8)$$

Much like the definition of $\bar{\Gamma}$ in the Farhadifar model, we have

$$\bar{\Omega} = \frac{\Omega}{KA_0}.$$

We now calculate the force on each vertex resulting from this line elasticity term. The gradient of this energy term is given by

$$\begin{aligned} \nabla E_{\Omega} &= \nabla \sum_j \frac{\Omega}{2} \ell_{i,j}^2, \\ &= \sum_j \Omega \ell_{i,j} (\nabla \ell_{i,j}), \\ &= \sum_j \Omega \ell_{i,j} (-\hat{\ell}_{i,j}). \end{aligned}$$

Therefore, since $\mathbf{F}_{\Omega} = -\nabla E_{\Omega}$,

$$\mathbf{F}_{\Omega} = \sum_j \Omega \ell_{i,j} \hat{\ell}_{i,j}. \quad (9)$$

The direction of this force is acquired from the calculations in Appendix A.

2.4 Simulating Cell Sheets

We used the python package `tyssue` [9] to obtain the data for this experiment. Tyssue is an open-source library which includes a range of vertex modelling tools. Importantly, it supports various energy functions and parameters which could be easily fine-tuned to match our requirements.

We see from the definitions in Equation (6) that many combinations of the four dimensional parameters result in the same values for $\bar{\Lambda}$ and $\bar{\Gamma}$. To avoid repeating simulations with identical values for the dimensionless parameters, all simulations will be done with $K = 1$ and $A_0 = 1$. Then only Λ and Γ determine the packing geometry, and they are equal to their nondimensionalised values.

Before we study the characteristics of epithelial sheets, we require a method for generating them. To do this, we first generate a regular hexagonal lattice which is 20 cells wide in both directions. We then pick a random cell to divide, and split it at a random angle. The random angle is chosen with a uniform distribution. All non-boundary cells also have an equal probability of dividing. This interior cell criterion is implemented to reduce the sheet overlapping itself. This nuance is discussed further in Section 4. We relax the sheet using a quasi-static solver with the desired model and parameters. The quasi-static solver performs gradient descent on the energy function with the BFGS algorithm [9]. This produces a realistic arrangement of cells, consistent with the model used for the simulation. Repeating the splitting process for 1000 iterations yields a disordered sheet with properties determined by the model. This whole process mimics the process of real tissue growth and mitosis [1].

The simulations were performed with a *free* boundary condition. By this, we mean that the vertices on the boundary were allowed to move freely. Their motion was entirely determined by the forces from the vertex model. This does come with some complications which are discussed in Section 4.

We tested the Farhadifar model with three different values for $\bar{\Gamma}$: 0.05, 0.1, and 0.2. For each of these values, a suitable range for $\bar{\Lambda}$ that produced stable tissues was determined. This range was determined experimentally by applying the quasi-static solver to a hexagonal lattice. For each value of $\bar{\Lambda}$, we observed whether the tissue collapsed and used these results to determine the range. We then simulated the growth of the sheet with a range of values for $\bar{\Lambda}$ in this interval. The line elasticity model was similarly tested with three values for $\bar{\Omega}$: 0.05, 0.1, 0.2. Each were tested on a suitable range for $\bar{\Lambda}$.

2.5 Validation

Before we used `tyssue` to study Lewis's law, we had to ensure its results were consistent with both models. To verify this, we first generated an irregular sheet of cells. Using equation (4), we picked a random vertex and calculated the force we expect from the Farhadifar Model. This was performed multiple times on different sheets with both boundary and interior vertices. We also performed the same analysis for the line elasticity model using equation (9).

An example of each of these tests is shown in Table 1. The sheet used for this example is displayed in Figure 3. In this example, we set $\bar{\Lambda} = 0.12$ and $\bar{\Gamma} = 0.04$

Farhadifar Model		
Direction	x	y
Expected Force	-6.7469596	1.66098053
<code>tyssue</code> Output	-6.746960	1.660981

Line Elasticity Model		
Direction	x	y
Expected Force	-5.90254711	1.54918789
<code>tyssue</code> Output	-5.902547	-1.549188

Table 1: The calculated force on the vertex displayed in Figure 3. The values calculated by `tyssue` match the expected results to 6 decimal places. This is true for both the Farhadifar model and line elasticity model.

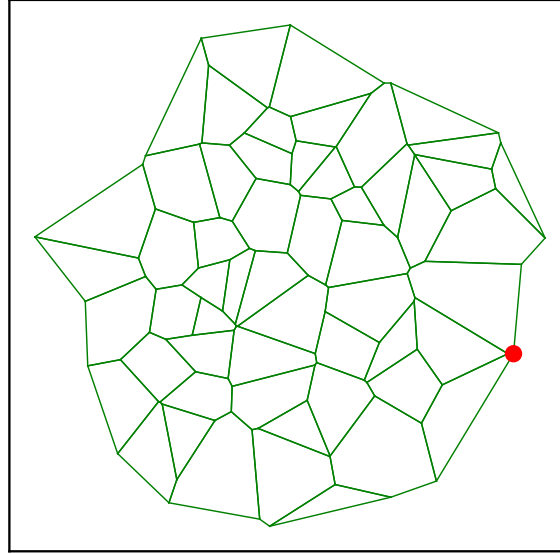


Figure 3: The sheet from which the forces in Table 1 were calculated. The vertex which experiences these forces is shown in red.

3 Results

3.1 Farhadifar Model

We now outline the full analysis performed on each set of parameters. We will take $\bar{\Gamma} = 0.1$, $\bar{\Lambda} = -0.4$ as an example. The tissue resulting from this simulation is shown in Figure 4A. Some sheets simulated with other parameters are shown in Appendix B. To verify the sheet has been disordered sufficiently, we analyse the proportion of cells with a given number of sides. A plot of these proportions are also shown in Figure 4B. From this, we see that these ratios are roughly constant for at least the last 500 iterations. This suggests the simulation length is appropriate.

Now that we have confirmed the validity of the simulation, we can turn our attention to Lewis's law. We plot the mean areas of cells with a given number of sides in Figure 4C. We also use a linear least squares regression to compute a line of best fit. This is of the form

$$A_{\text{avg}} = m \cdot n + d, \quad (10)$$

where A_{avg} is the average area of the set of cells with n sides. As expected, m is positive. This means cells with a greater number of sides tend to be larger.

Of the two parameters, m and d , we are mainly interested in the gradient, m . We obtain this value of m for a variety of $\bar{\Gamma}$ and $\bar{\Lambda}$ values and plot these in Figure 4D. This shows that increasing $\bar{\Gamma}$ or $\bar{\Lambda}$ both increase the gradient m . This demonstrates that higher tensions produce a greater disparity in the mean cell area. When these values are small, the gradient approaches 0. This is the regime in which energy minimisation is dominated by the area elasticity term. In this scenario, all cells tend to have an area close to A_0 and the length of the perimeter is less important. This matches the results from Farhadifar et al. (2007) [1] and is what we would expect based on Figure 2.

Upon inspection of these results, we notice that they follow a logarithmic curve described by

$$m = a \ln(b - \bar{\Lambda}) + c. \quad (11)$$

Here, a , b , and c are parameters of this curve which depend on $\bar{\Gamma}$. These values are determined with a least squares regression. The values of these parameters are presented in Table 2.

$\bar{\Gamma}$	a	b	c
0.05	-0.092 ± 0.015	0.203 ± 0.002	-0.034 ± 0.014
0.1	-0.122 ± 0.003	0.057 ± 0.004	-0.0240 ± 0.0016
0.2	-0.16 ± 0.04	-0.01 ± 0.11	0.051 ± 0.016

Table 2: The fitted parameters for equation (11) from the Farhadifar model.

Out of all three of the parameters in equation (11), the most interesting is b . This represents the value of $\bar{\Lambda}$ at which the gradient diverges to infinity. We hypothesise that this corresponds to the invalid phase region in Figure 2. By comparing the results from Table 2 with Figure 2 we see that these results appear to match.

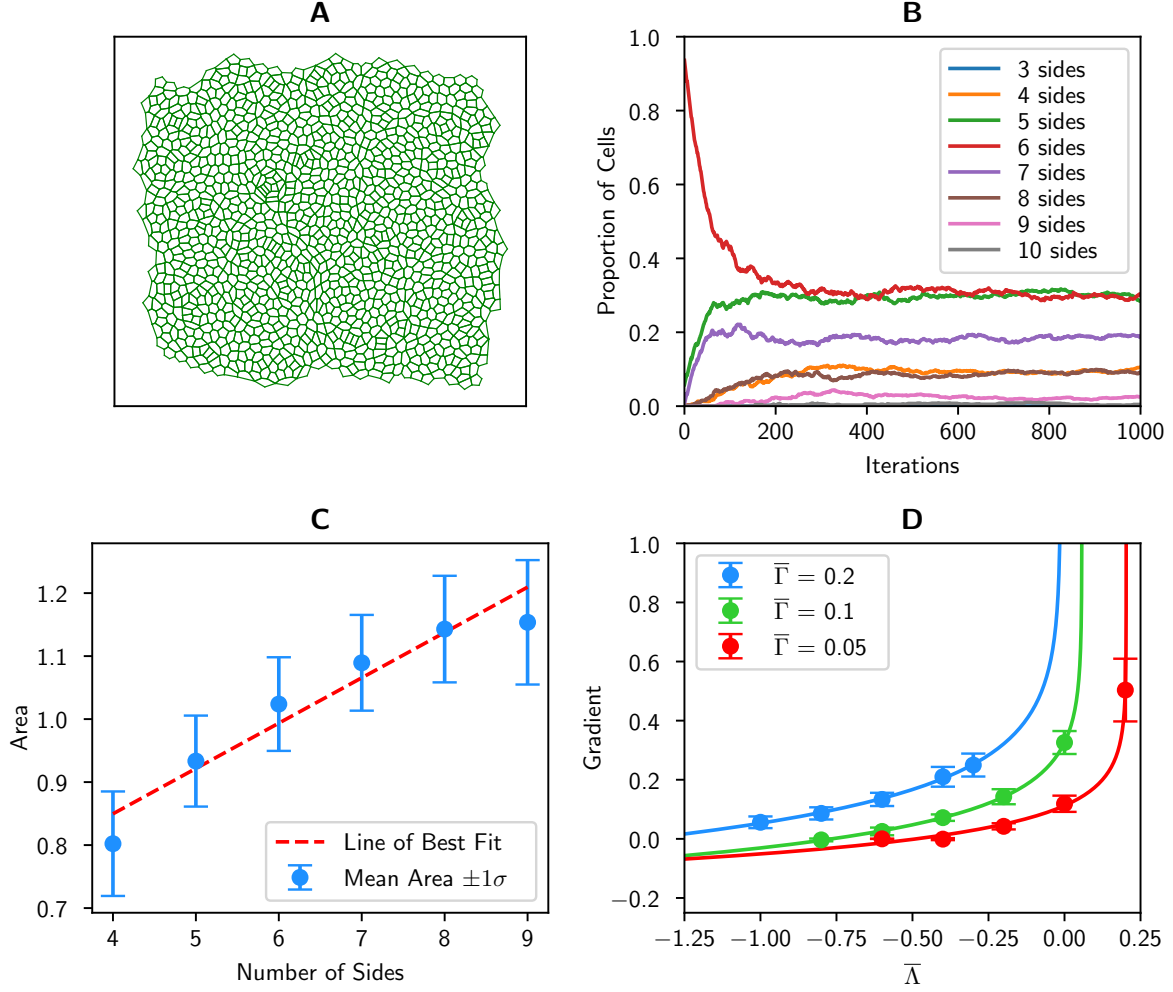


Figure 4: (A) The result of performing the tissue growth process outlined in Section 2.4 with $\bar{\Gamma} = 0.1$ and $\bar{\Lambda} = -0.4$. (B) The proportion of cells with a given number of sides throughout the growth process of the $\bar{\Gamma} = 0.1$, $\bar{\Lambda} = -0.4$ sheet. The proportions are roughly constant at the end of the simulation, indicating a sufficient number of iterations were completed. (C) The mean area of the set of n -sided cells in Figure 4A with a line of best fit. The error bars are the standard deviations in the area of the sets of n -sided cells. The red dashed line is the line of best fit which was computed with the least squares method. This represents the Lewis's law relationship for the sheet. (D) The gradient of the Lewis's law relationship for various combinations of $\bar{\Gamma}$ and $\bar{\Lambda}$. The error bars are the uncertainty in the gradient for a given set of parameters. These are obtained by taking the square root of the diagonal elements of the covariance matrix obtained after finding the line of best fit for each set of parameters. The curves are Equation (11) fitted for each value of $\bar{\Gamma}$ using the least squares method.

3.2 Line Elasticity Model

We now perform the same analysis on the line elasticity model. As an example, we take $\bar{\Omega} = 0.1$ and $\bar{\Lambda} = 0.1$. The result of this simulation is shown in Figure 5A. Similar visualisations for some other parameters are shown in Appendix C. A plot of the proportion of cells with a given number of sides is shown in Figure 5B.

Crucially, we observe that the simulated sheet contains many thin, elongated cells. These do not appear in simulations for the Farhadifar model. This is because the perimeter of these long cells is relatively large. In a sheet governed by the Farhadifar model, these cells would experience large contractile forces. However, these cells have a larger number of sides, and so the individual side lengths are of a comparable length to other, smaller cells in the sheet. In the line elasticity model, we are only concerned with individual side length, not cell perimeter, and so we observe the existence of these long, thin cells.

The Lewis's law relationship for this example is plotted in Figure 5C and the gradient for every set of parameters is shown in Figure 5D. In general, the results in Figure 5D are similar to those in Figure 4D. Larger tensions

produce a greater disparity in mean areas. We also observe a regime for small $\bar{\Lambda}$ and $\bar{\Gamma}$ where the area elasticity term dominates energy minimisation. Finally, the best fit parameters for equation (11) are shown in Table 3.

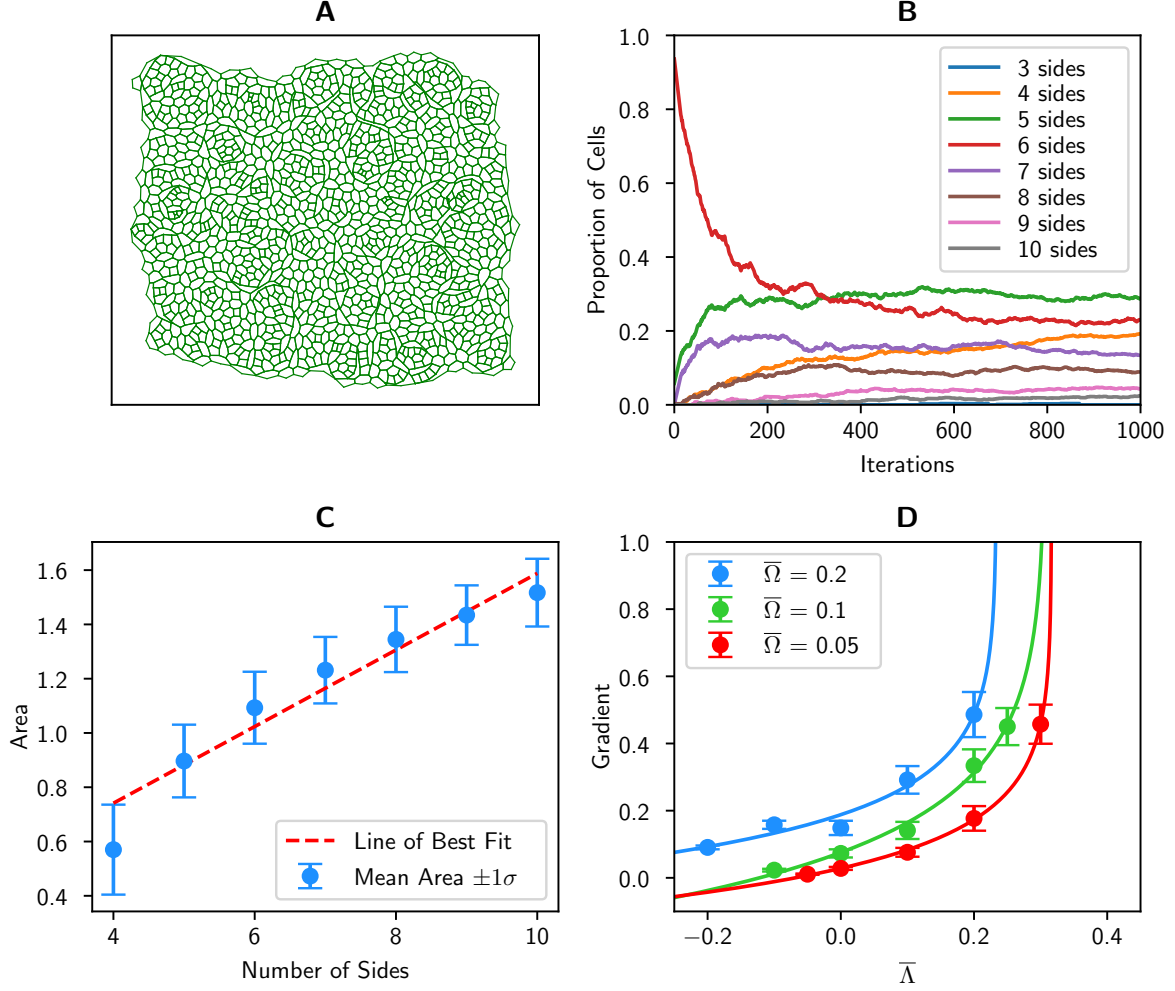


Figure 5: (A) The result of performing the tissue growth process outlined in Section 2.4 with $\bar{\Omega} = 0.1$ and $\bar{\Lambda} = 0.1$. (B) The proportion of cells with a given number of sides throughout the growth process of the $\bar{\Omega} = 0.1$, $\bar{\Lambda} = 0.1$ sheet. The proportions are roughly constant at the end of the simulation, indicating a sufficient number of iterations were completed. (C) The mean area of the set of n -sided cells in Figure 5A with a line of best fit. The error bars are the standard deviations in the area of the sets of n -sided cells. The red dashed line is the line of best fit which was computed with the least squares method. This represents the Lewis's law relationship for the sheet. (D) The gradient of the Lewis's law relationship for various combinations of $\bar{\Omega}$ and $\bar{\Lambda}$. The error bars are the uncertainty in the gradient for a given set of parameters. These are obtained by taking the square root of the diagonal elements of the covariance matrix obtained after finding the line of best fit for each set of parameters. The curves are Equation (11) fitted for each value of $\bar{\Omega}$ using the least squares method.

$\bar{\Omega}$	a	b	c
0.05	-0.145 ± 0.010	0.316 ± 0.003	-0.139 ± 0.012
0.1	-0.23 ± 0.06	0.31 ± 0.03	-0.19 ± 0.05
0.2	-0.15 ± 0.06	0.23 ± 0.04	-0.04 ± 0.06

Table 3: The fitted parameters for equation (11) from the line elasticity model.

As noted in Section 3.1, the parameter b is believed to represent the point at which a tissue becomes unstable. The corresponding values for b in the line elasticity model could therefore aid in finding a similar phase plot to 2 for this model.

4 Conclusions and Discussion

We successfully recreated the results from Farhadifar et al. [1]. The gradient of the Lewis's law relationship depends on the nondimensionalised parameters of the Farhadifar model. Furthermore, we found that the effect $\bar{\Lambda}$ has on the gradient can be described by Equation (11). We successfully fit this curve to our results and obtained clear trends in the parameters a , b , and c . We also showed that Lewis's law holds for a line elasticity model described by equation (7). Equation (11) describes the effect of $\bar{\Lambda}$ on the gradient of Lewis's law for this model as well.

A key observation of the results is that the mean area of four-sided cells appears smaller than we would expect from Lewis's law. This is true for both the Farhadifar and line-elasticity models. One possible reason for this is the tolerances used in the quasi-static solver. In preliminary simulations, the tolerances were far too high. This meant that the splitting of small four-sided cells did not induce large enough forces on any vertices to cause a rearrangement of the sheet. For the full simulations used in this report, the tolerances were reduced such that the first 100 iterations all resulted in a new arrangement of vertices. This meant that for the gradient descent process to terminate, each component of the gradient on all the vertices must be less than 10^{-7} . Furthermore, the change in the total sheet energy must also be less than 10^{-7} between iterations of the BFGS algorithm. The total energy stored in a sheet the size used in these simulations is typically of order 10^2 . As seen in Table 1, forces are typically of order unity. For these reasons, it was expected that the chosen tolerances would be sufficient. However, if the tolerances were still too high, this would account for the small area of four-sided cells in the simulations. Further testing with smaller tolerances would be required to see if this is the case.

As discussed in Section 3.1, the form of Equation (11) was not derived from the physics of the system. Rather, it was chosen because it seemed to successfully fit the Farhadifar model data in Figure 4D. Further investigation is required to discover why this model appears to work so well. Notably, when fitting this model to the line elasticity results, it appears as though the $\bar{\Omega} = 0.05$ and $\bar{\Omega} = 0.1$ curves intersect. It would be surprising if this were the case. This would imply that for some $\bar{\Lambda}$, the same packing geometry can be replicated with two unique values for $\bar{\Omega}$. Testing over a wider range of parameters would be the first step in investigating this result further. We would also like to investigate the effect $\bar{\Gamma}$ and $\bar{\Omega}$ have on the parameters in Equation (11). Testing more $\bar{\Gamma}$ and $\bar{\Omega}$ values would illuminate trends in the model parameters a , b , and c .

One notable difference from our report and the work of Farhadifar et al. [1] is the boundary conditions. In their simulations, they used what is known as *periodic* boundary condition. This is a technique in which the forces on the left boundary of the sheet are applied to the right boundary and vice-versa. The same is done for the top and bottom boundaries. In this way, the sheet essentially tessellates the plane and mimics a very large sheet of cells. In our simulations we used a free boundary condition. This simply allows the vertices to move freely without additional forces or restrictions. A consequence of this method is that the sheet had a tendency to overlap at the boundaries. This occurred more frequently for very large or very small values of $\bar{\Lambda}$, $\bar{\Gamma}$, and $\bar{\Omega}$. In future research, implementing a collision detection library would allow a wider range of parameters to be analysed. Alternatively, utilising a periodic boundary condition would avoid this complication and improve the efficiency of the simulations.

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Appendix

A Forces in the Farhadifar Model

Here we outline the process for computing the directions of the forces in the Farhadifar model. We begin with the area elasticity force. For this component we can limit our analysis to only the triangle formed by vertex i and its two adjacent vertices. This is represented by the grey region in Figure 6. The remaining cell area is constant with respect to $\mathbf{R}_i$, and thus does not affect the gradient calculation. The area of this triangle is given by

$$A_{\Delta} = \frac{1}{2}bh.$$

Here, b and h are simply the base and height of the triangle, and we define new coordinate axes by the directions of these lines. We denote these coordinates of $\mathbf{R}_i$ as x_b and x_h . The axes are orthogonal and shown in Figure 6. We now compute the gradient with respect to our choice in coordinates using

$$\begin{aligned}\nabla A_{\alpha} &= \left(\frac{\partial A_{\Delta}}{\partial x_b}, \frac{\partial A_{\Delta}}{\partial x_h} \right), \\ &= \left(0, \frac{b}{2} \right).\end{aligned}$$

Thus, we have our expression for ∇A_{α} . This is more clearly expressed as $-\frac{b}{2}\hat{\mathbf{n}}_b$ where $\hat{\mathbf{n}}_b$ is the unit vector pointing to the interior of the cell, normal to the base of the triangle. This is the same vector as shown in Figure 1.

We now discuss the direction of the line tension forces. As with the area elasticity term, we will choose new coordinate axes which simplify the calculation. We choose x_{ℓ} to be the coordinate of $\mathbf{R}_i$ in the direction of the edge and denote the orthogonal coordinate as x_n . Then the gradient of the length of an edge is given by

$$\begin{aligned}\nabla \ell_{i,j} &= \left(\frac{\partial \ell_{i,j}}{\partial x_{\ell}}, \frac{\partial \ell_{i,j}}{\partial x_n} \right), \\ &= (-1, 0)\end{aligned}$$

This is simply the negative of the unit vector from vertex i to j which we call $-\hat{\ell}_{i,j}$. Finally, we note that the edge lengths of the cell are constant with respect to $\mathbf{R}_i$ unless they are adjacent to vertex i . Therefore, the gradient of the perimeter is simply the sum of the gradients of only those edges adjacent to vertex i . As we now know the gradient of the edge lengths, this yields

$$\nabla P_{\alpha} = -\hat{\ell}_{i,1} - \hat{\ell}_{i,2},$$

where vertices 1 and 2 are adjacent to vertex i .

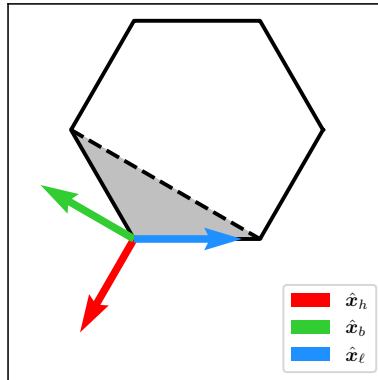


Figure 6: The directions relevant for finding the forces in the Farhadifar model. The grey region represents an area of the cell which is not constant with respect to $\mathbf{R}_i$.

B Farhadifar Simulated Tissues

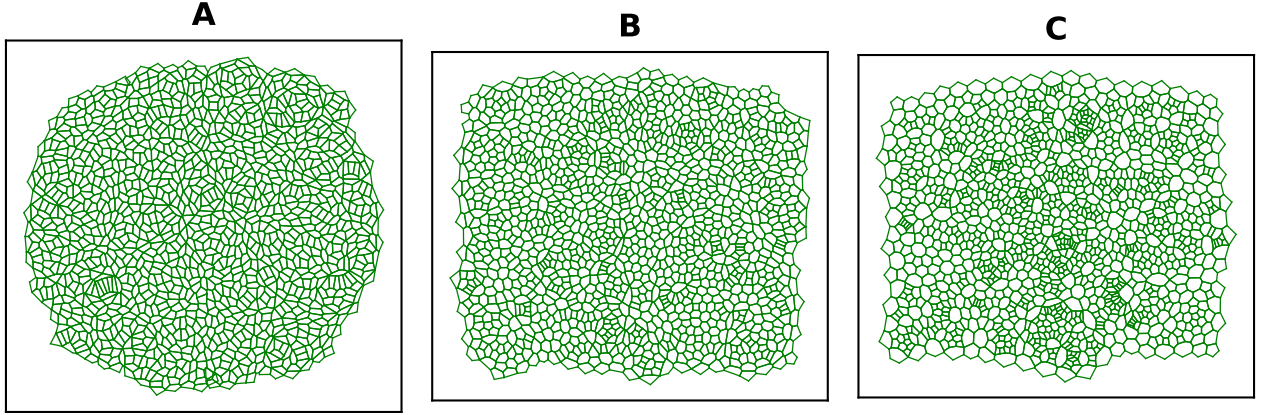


Figure 7: The results of the tissue growth process for three different sets of parameters for the Farhadifar model. (A) $\bar{\Gamma} = 0.05$ and $\bar{\Lambda} = -0.4$. (B) $\bar{\Gamma} = 0.1$ and $\bar{\Lambda} = -0.2$. (C) $\bar{\Gamma} = 0.2$ and $\bar{\Lambda} = -0.2$.

C Line Elasticity Simulated Tissues

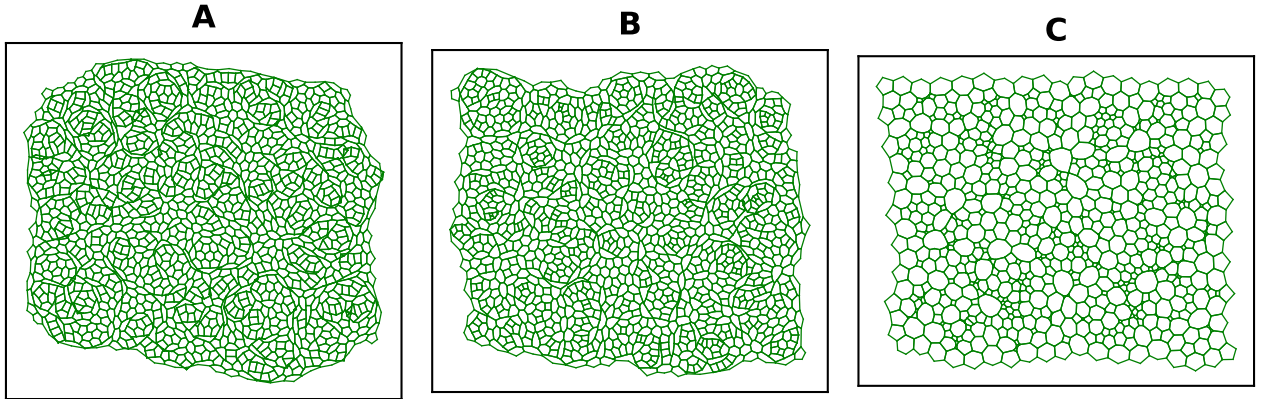


Figure 8: The results of the tissue growth process for three different sets of parameters for the line elasticity model. (A) $\bar{\Omega} = 0.05$ and $\bar{\Lambda} = 0$. (B) $\bar{\Omega} = 0.1$ and $\bar{\Lambda} = 0.1$. (C) $\bar{\Omega} = 0.2$ and $\bar{\Lambda} = 0.2$.

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